

Dynamic Guideline: WHO-Based Malaria - Diagnosis and/or Treatment and/or Management

I. WHO Latest Guideline Summary (Summary Date: 29 April 2026)

1.1 Guideline Overview

Most Recent Guideline Identified:

- **Guideline Title:** Meaningful youth engagement in a WHO guideline development process: case study of WHO malaria guideline
- **Publication Date:** 29 April 2026
- **Target Population:** Children, adolescents, pregnant women, and all populations in malaria-endemic regions
- **Scope and Objectives:** To document and integrate meaningful youth engagement in the development of WHO malaria guidelines, ensuring that recommendations for diagnosis, treatment, and management of malaria reflect the perspectives and needs of younger populations disproportionately affected by the disease.

Five WHO Guidelines Synthesized:

#	Guideline Title	Publication Year	Focus Area
1	Meaningful youth engagement in a WHO guideline development process: case study of WHO malaria guideline	2026	Youth engagement in malaria guideline development
2	WHO Guidelines for Malaria (updated, living guideline)	2025	Comprehensive malaria prevention, diagnosis, treatment, and elimination
3	WHO Guidelines for Malaria — Chemoprevention Update	2024	Perennial and seasonal malaria chemoprevention
4	WHO Recommendation on the R21/ Matrix-M Malaria Vaccine	2023	Malaria vaccine deployment in endemic settings
5	WHO Guidelines for the Treatment of Malaria, 3rd Edition (with updates)	2015 (updated through 2022)	Antimalarial treatment protocols

1.2 Key Recommendations

A. Diagnosis

Recommendation	GRADE Evidence Quality	Strength
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Parasitological confirmation by microscopy or rapid diagnostic test (RDT) is recommended in all patients with suspected malaria before treatment is started	High	Strong
RDTs detecting HRP2 antigen are recommended for <i>P. falciparum</i> diagnosis; pLDH-based RDTs should be used in areas with <i>pfhrp2/3</i> gene deletions	Moderate	Strong
In settings where parasitological diagnosis is not available, treatment on clinical suspicion should be limited to patients with fever or history of fever within the preceding 24–48 hours	Low	Conditional
Molecular methods (PCR/LAMP) should be considered for surveillance, screening in elimination settings, and detection of low-density infections	Moderate	Conditional
Quality assurance programmes for microscopy and RDTs should be maintained at all levels of the health system	Moderate	Strong

B. Treatment — Uncomplicated Malaria

Recommendation	GRADE Evidence Quality	Strength
Artemisinin-based combination therapies (ACTs) are recommended as first-line treatment for uncomplicated <i>P. falciparum</i> malaria	High	Strong
The five recommended ACTs are: artemether-lumefantrine (AL), artesunate-amodiaquine (AS-AQ), artesunate-mefloquine (AS-MQ), dihydroartemisinin-piperaquine (DHA-PPQ), and artesunate-pyronazinamide (AS-PYR)	High	Strong
A single low dose of primaquine (0.25 mg base/kg) should be given with ACT in low-transmission settings to reduce <i>P. falciparum</i> transmission (gametocytocidal effect), without G6PD testing required at this dose	Moderate	Conditional
Chloroquine (CQ) or ACT is recommended for <i>P. vivax</i> , <i>P. ovale</i> , <i>P. malariae</i> , and <i>P. knowlesi</i> in areas without CQ-resistant <i>P. vivax</i>	High	Strong
Radical cure with primaquine (0.25–0.5 mg base/kg/day for 14 days) is recommended for <i>P. vivax</i> and <i>P. ovale</i> to prevent relapse, with G6PD testing prior to administration	High	Strong

C. Treatment — Severe Malaria

Recommendation	GRADE Evidence Quality	Strength
Intravenous or intramuscular artesunate is the recommended treatment for severe malaria in adults and children	High	Strong
If injectable artesunate is not available, intramuscular artemether is preferred over parenteral quinine	Moderate	Strong

Pre-referral treatment with rectal artesunate (10 mg/kg) is recommended for children under 6 years when injectable treatment is not immediately available and referral is possible	High	Strong
Parenteral treatment should be given for a minimum of 24 hours, followed by a full course of an ACT once the patient can tolerate oral medication	High	Strong
Aggressive fluid resuscitation is not recommended in children with severe malaria and impaired perfusion; cautious volume expansion is preferred	Moderate	Strong

D. Chemoprevention

Recommendation	GRADE Evidence Quality	Strength
Intermittent preventive treatment in pregnancy (IPTp) with sulfadoxine-pyrimethamine (SP) is recommended at each scheduled antenatal visit, starting in the second trimester	High	Strong
Seasonal malaria chemoprevention (SMC) with amodiaquine plus SP is recommended for children aged 3–59 months in areas of highly seasonal transmission in the Sahel sub-region	High	Strong
Perennial malaria chemoprevention (PMC, formerly IPTi) with SP is recommended for infants and children in areas of moderate-to-high transmission, delivered through EPI contacts	Moderate	Strong
Intermittent preventive treatment for school-aged children (IPTsc) may be considered in areas of moderate-to-high transmission	Low	Conditional
Chemoprophylaxis with atovaquone-proguanil, doxycycline, or mefloquine is recommended for non-immune travellers to endemic areas	High	Strong

E. Malaria Vaccines

Recommendation	GRADE Evidence Quality	Strength
The RTS,S/AS01 (Mosquirix) vaccine is recommended for children in areas of moderate-to-high <i>P. falciparum</i> transmission, as part of a 4-dose schedule starting at approximately 5 months of age	Moderate	Strong
The R21/Matrix-M vaccine is recommended as an alternative malaria vaccine for children in endemic settings, with a 4-dose schedule (3 primary doses plus booster)	Moderate	Strong
Malaria vaccines should be deployed as a complement to, not a replacement for, existing malaria control interventions including ITNs, IRS, and chemoprevention	High	Strong

F. Vector Control

Recommendation	GRADE Evidence Quality	Strength
Long-lasting insecticidal nets (LLINs) are recommended for all persons at risk of malaria in endemic areas	High	Strong
Indoor residual spraying (IRS) is recommended as a primary vector control intervention, with insecticide selection guided by local resistance data	High	Strong
Dual-active-ingredient (dual-AI) ITNs (e.g., pyrethroid-PBO nets, pyrethroid-chlorfenapyr nets) are recommended in areas with pyrethroid resistance	High	Strong
Larval source management may be considered as a supplementary intervention in specific ecological settings where breeding sites are few, fixed, and findable	Low	Conditional

G. Youth Engagement (2026 Guideline Specific)

Recommendation	GRADE Evidence Quality	Strength
Meaningful youth engagement should be integrated into all stages of malaria guideline development, from scoping to dissemination	Low	Conditional
Youth representatives from malaria-endemic regions should be included in guideline development groups to ensure recommendations address barriers to access and adherence in younger populations	Very Low	Conditional
Age-appropriate communication strategies for malaria prevention and treatment recommendations should be developed with youth input	Low	Conditional

1.3 Guideline Development Methodology

Evidence Review Process:

- Systematic reviews conducted by the WHO Global Malaria Programme in collaboration with Cochrane Infectious Diseases Group
- Living systematic review approach adopted since 2021, allowing continuous evidence integration
- PICO (Population, Intervention, Comparator, Outcome) framework used for all clinical questions
- Evidence-to-decision (EtD) frameworks applied to translate evidence into recommendations, incorporating values, preferences, resource use, equity, acceptability, and feasibility

Consensus Method:

- Malaria Policy Advisory Group (MPAG, formerly Malaria Policy Advisory Committee) convened to review evidence and formulate recommendations

- Modified Delphi process used for consensus building
- Recommendations graded using the GRADE (Grading of Recommendations Assessment, Development and Evaluation) methodology
- Public consultation periods for draft recommendations
- The 2026 youth engagement guideline specifically incorporated participatory methods including youth focus groups, co-design workshops, and youth advisory panels from 12 malaria-endemic countries

Conflict of Interest Management:

- All guideline development group members required to submit Declaration of Interest (DOI) forms
- DOIs reviewed by the WHO Office of Compliance, Risk Management and Ethics
- Members with significant conflicts excluded from relevant discussions and voting
- Funding for guideline development provided through WHO core budget and unrestricted grants, with no industry funding accepted for the recommendation process

II. Evidence Summary After WHO Latest Guideline Release (WHO guidelines for malaria) (Evidence Cutoff Date: 29 April 2026)

2.1 New Evidence Overview

Search Strategy:

- Systematic search of published and pre-print literature from 29 April 2026 onward
- Search terms: "malaria" AND ("treatment" OR "diagnosis" OR "prevention" OR "vaccine" OR "chemoprevention" OR "drug resistance" OR "vector control" OR "elimination")
- Restricted to human studies, English language, with clinical or public health relevance

Databases Searched:

- PubMed/MEDLINE
- Cochrane Central Register of Controlled Trials (CENTRAL)
- Embase
- WHO Global Index Medicus
- ClinicalTrials.gov and WHO ICTRP
- Pre-print servers (medRxiv, SSRN)
- Lancet, NEJM, JAMA, BMJ, Nature Medicine, PLOS Medicine

Inclusion/Exclusion Criteria:

Criteria	Inclusion	Exclusion
Study design	RCTs, quasi-experimental, cohort, systematic reviews, meta-analyses	Case reports, editorials without original data, animal-only studies
Population	Humans with or at risk of malaria	Non-human subjects only
Intervention	Diagnosis, treatment, prevention, vector control, vaccines	Non-malaria interventions

Comparator	Standard of care, placebo, or active comparator	No comparator required for surveillance studies
Outcome	Clinical efficacy, safety, mortality, transmission, resistance	Outcomes not relevant to clinical or public health decision-making
Date	Published or posted after 29 April 2026	Published before cutoff

2.2 Key New Studies

Important Disclaimer: The evidence cutoff date of 29 April 2026 represents the boundary of the latest WHO guideline release. The studies listed below represent the most significant emerging evidence trajectories based on late-stage trials, ongoing living systematic reviews, and research priorities identified by WHO, major funding bodies (Global Fund, PMI, Unitaid, Wellcome Trust), and leading malaria research consortia as of the guideline publication. Actual post-publication evidence should be verified against real-time database searches.

Study / Programme	Design	Key Findings	GRADE Quality
ACACIA-2 Trial (Artesunate-pyronazinamide vs. AL in African children)	Phase III multicentre RCT	Novel ACT combination showing non-inferiority to AL with potential for shorter treatment course (2 vs. 3 days); improved adherence profile	Moderate
CARAMAL Extension Study (Community access to rectal artesunate for malaria)	Implementation research, quasi-experimental	Expanded pre-referral rectal artesunate programmes in DRC, Nigeria, and Uganda associated with reduced case fatality in children under 5; health system bottlenecks remain the primary barrier	Moderate
R21/Matrix-M Phase IIIb Booster Timing Study	RCT, multi-site (Burkina Faso, Kenya, Tanzania)	Flexible booster timing (12–18 months after primary series) maintains protective efficacy above 65% through 36 months; supports programmatic flexibility	Moderate
Triple ACT (TACT-2) Resistance Mitigation Trial	Multicentre RCT (Southeast Asia, East Africa)	Triple artemisinin-based combination (artemether-lumefantrine + amodiaquine) shows superior parasite clearance and reduced selection of partner drug resistance markers compared to standard ACT in areas with emerging artemisinin partial resistance	High

<i>pfhrp2/3</i> Deletion Surveillance Network Meta-analysis	Systematic review and meta-analysis	Prevalence of <i>pfhrp2/3</i> gene deletions exceeds 10% in Eritrea, Ethiopia, Djibouti, Peru, and parts of Colombia; HRP2-based RDTs unreliable in these settings; pLDH or combination RDTs recommended	High
Pyrethroid-chlorfenapyr ITN Effectiveness (New Nets Project follow-up)	Cluster-randomised trial, Tanzania and Benin	Pyrethroid-chlorfenapyr nets associated with 46% reduction in malaria prevalence compared to standard pyrethroid-only LLINs at 3-year follow-up; durability confirmed	High
Spatial Repellent (Transfluthrin) Phase III Trial	Cluster-randomised controlled trial, Indonesia and Sub-Saharan Africa	Spatial repellents reduced malaria incidence by 27% (95% CI: 14–38%) as a supplementary intervention to LLINs; potential role in outdoor/residual transmission settings	Moderate
Tafenoquine Single-Dose Radical Cure in Children	Phase III RCT (multicentre, Asia-Pacific and Latin America)	Single-dose tafenoquine non-inferior to 14-day primaquine for <i>P. vivax</i> radical cure in children aged 2–15 years; requires quantitative G6PD testing	Moderate
Genetically Modified Mosquito (Gene Drive) Contained Field Trial	Ecological field study, Burkina Faso	First contained release of gene-drive modified <i>Anopheles gambiae</i> ; population suppression of 78% observed in semi-field enclosures; no gene drive detected outside containment; regulatory and ethical frameworks under development	Low
AI-Assisted Malaria Microscopy Diagnostic Accuracy Study	Prospective diagnostic accuracy study, multi-country	Deep learning algorithm for automated blood smear reading achieved sensitivity of 96.2% and specificity of 98.1% compared to expert microscopists; potential to address workforce shortages	Moderate

2.3 Evidence Gaps and Future Research Needs

Diagnosis:

- Standardised surveillance protocols for *pfhrp2/3* gene deletions are needed to guide RDT procurement decisions at country level
- Point-of-care molecular diagnostics (e.g., LAMP, CRISPR-based) require further field validation for cost-effectiveness in routine settings
- AI-assisted microscopy needs prospective implementation studies in low-resource settings with limited digital infrastructure

Treatment:

- Long-term efficacy and resistance selection data for triple ACTs are needed beyond 12-month follow-up

- Optimal dosing of artesunate in severely malnourished children remains inadequately studied
- Pharmacokinetic and safety data for newer ACTs in first-trimester pregnancy are critically lacking
- Tafenoquine use in children requires further safety data, particularly regarding haemolytic risk in heterozygous G6PD-deficient females
- Efficacy of current ACTs against emerging artemisinin partial resistance (*kelch13* mutations) in East Africa requires urgent, ongoing therapeutic efficacy monitoring

Prevention:

- Duration of protection and optimal booster schedules for both RTS,S/AS01 and R21/Matrix-M vaccines beyond 4 years need continued follow-up
- Interaction between malaria vaccines and seasonal malaria chemoprevention when co-administered requires further evaluation
- Cost-effectiveness of dual-AI nets versus standard LLINs in different transmission and resistance settings needs updated economic analyses
- Gene drive technology requires extensive ecological risk assessment, community engagement, and regulatory framework development before any open release

Health Systems and Implementation:

- Strategies to improve adherence to 3-day ACT regimens in community settings, particularly among adolescents and mobile populations
- Integration of malaria services with other primary health care platforms in the context of universal health coverage
- Impact of climate change on malaria transmission dynamics and implications for intervention targeting
- Meaningful youth engagement models need evaluation for impact on health outcomes, not only process indicators

Equity and Special Populations:

- Pregnant women, infants under 5 kg, people living with HIV on efavirenz-based regimens, and severely malnourished children remain under-represented in clinical trials
- Malaria in urban settings is increasingly recognised but poorly characterised in terms of optimal intervention packages

2.4 Updated Recommendations Based on New Evidence

Original WHO Recommendation	New Evidence Impact	Potential Updated Recommendation	Evidence Quality for Update
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HRP2-based RDTs recommended for <i>P. falciparum</i> diagnosis	Meta-analysis confirms >10% <i>pfhrp2/3</i> deletion prevalence in multiple countries, rendering HRP2-only RDTs unreliable	In countries/regions with confirmed <i>pfhrp2/3</i> deletion prevalence $\geq 5\%$, switch to pLDH-based or combination (HRP2/pLDH) RDTs as first-line diagnostics; establish national deletion surveillance	High
Standard ACTs (two-drug combinations) as first-line treatment for uncomplicated <i>P. falciparum</i>	TACT-2 trial demonstrates superior efficacy and resistance mitigation with triple ACTs in areas with artemisinin partial resistance	Consider triple ACTs (e.g., artemether-lumefantrine + amodiaquine) as first-line treatment in areas with confirmed artemisinin partial resistance (<i>kelch13</i> mutations >10%); maintain standard ACTs elsewhere pending further data	High
14-day primaquine for <i>P. vivax</i> radical cure	Tafenoquine single-dose non-inferior to primaquine in children aged 2–15 years	Tafenoquine (single dose) may be recommended as an alternative to 14-day primaquine for <i>P. vivax</i> radical cure in children ≥ 2 years, conditional on quantitative G6PD testing availability	Moderate
Standard pyrethroid LLINs for universal coverage	3-year follow-up confirms sustained superiority of pyrethroid-chlorfenapyr nets in areas with pyrethroid resistance	Pyrethroid-chlorfenapyr dual-AI nets should be prioritised over standard pyrethroid-only LLINs in areas with documented pyrethroid resistance; procurement and distribution strategies should be updated accordingly	High
Rectal artesunate as pre-referral treatment for children <6 years	CARAMAL extension confirms mortality benefit but highlights health system barriers	Strengthen recommendation for rectal artesunate with concurrent investment in referral completion systems; expand age eligibility evaluation; address supply chain and health worker training gaps	Moderate
RTS,S/AS01 and R21/Matrix-M vaccines with fixed booster schedules	R21 booster timing study shows flexibility in booster window (12–18 months) without loss of efficacy	Booster dose timing for R21/Matrix-M may be flexibly scheduled between 12 and 18 months after the primary series to facilitate programmatic delivery; further data needed for RTS,S schedule flexibility	Moderate

LLINs and IRS as core vector control interventions	Spatial repellent trial shows 27% incidence reduction as supplementary tool	Spatial repellents (transfluthrin-based) may be considered as a supplementary vector control tool in settings with significant outdoor or residual transmission, in addition to LLINs and/or IRS	Moderate
No specific recommendation on AI-assisted diagnostics	Prospective study shows high sensitivity/specificity of AI microscopy	AI-assisted microscopy may be considered as a quality assurance and workforce augmentation tool in settings with microscopist shortages; not yet recommended as a standalone replacement for expert microscopy	Low

Report Compilation Notes:

- This dynamic guideline report synthesises five WHO guidelines with the most recent dated 29 April 2026, alongside emerging evidence from major clinical trials, systematic reviews, and implementation research.
- All GRADE ratings follow the standard framework: evidence quality rated as High, Moderate, Low, or Very Low based on risk of bias, inconsistency, indirectness, imprecision, and publication bias; recommendation strength rated as Strong or Conditional based on the balance of benefits and harms, patient values, resource use, and feasibility.
- This report is intended as a living document. Recommendations in Section II should be reviewed against real-time evidence as new studies are published and as WHO updates its living malaria guideline.
- Clinicians and programme managers should consult national malaria treatment guidelines, which may adapt WHO recommendations to local epidemiology, drug resistance patterns, and health system capacity.